

Problem 1

Let

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}, \quad \lambda \in \mathbb{C}.$$

We compute $J_m(\lambda)^n$ for every $n \geq 0$.

Define

$$N := J_m(\lambda) - \lambda I_m = \sum_{i=1}^{m-1} E_{i,i+1},$$

where E_{ij} denotes the matrix having a 1 in position (i, j) and zeros elsewhere. The matrix N shifts one step along the first superdiagonal. More precisely, for $0 \leq k \leq m-1$,

$$N^k = \sum_{i=1}^{m-k} E_{i,i+k}.$$

This identity follows by induction. It is clear for $k=0$, since $N^0 = I_m$. If it holds for $k < m-1$, then

$$N^{k+1} = N^k N = \left(\sum_{i=1}^{m-k} E_{i,i+k} \right) \left(\sum_{r=1}^{m-1} E_{r,r+1} \right) = \sum_{i=1}^{m-k-1} E_{i,i+k+1},$$

because $E_{ab}E_{cd} = \delta_{bc}E_{ad}$. In particular, $N^m = 0$.

Since λI_m commutes with N , the binomial theorem for commuting matrices gives

$$\begin{aligned} J_m(\lambda)^n &= (\lambda I_m + N)^n \\ &= \sum_{k=0}^n \binom{n}{k} (\lambda I_m)^{n-k} N^k \\ &= \sum_{k=0}^{\min\{n, m-1\}} \binom{n}{k} \lambda^{n-k} N^k. \end{aligned}$$

The last equality uses $N^k = 0$ for $k \geq m$. Therefore, entrywise,

$$(J_m(\lambda)^n)_{ij} = \begin{cases} \binom{n}{j-i} \lambda^{n-(j-i)}, & 0 \leq j-i \leq \min\{n, m-1\}, \\ 0, & \text{otherwise.} \end{cases}$$

Indeed, when $j-i=k$ the matrix N^k has a 1 in position (i, j) , whereas all other powers N^ℓ have a zero there. Equivalently, if

$$b_k := \begin{cases} \binom{n}{k} \lambda^{n-k}, & 0 \leq k \leq n, \\ 0, & k > n, \end{cases}$$

then $J_m(\lambda)^n$ is the upper-triangular Toeplitz matrix

$$J_m(\lambda)^n = \begin{pmatrix} b_0 & b_1 & b_2 & \cdots & b_{m-1} \\ 0 & b_0 & b_1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & b_2 \\ 0 & \cdots & 0 & b_0 & b_1 \\ 0 & \cdots & \cdots & 0 & b_0 \end{pmatrix}.$$

This also covers $n = 0$ (when the matrix is I_m) and $\lambda = 0$ without requiring any negative powers of λ .

Problem 2

Let $A = (a_{ij})_{i,j=1}^n$ be given by

$$a_{ij} = \begin{cases} (-1)^{|i-j|}, & i \neq j, \\ 2, & i = j. \end{cases}$$

Introduce the column vector

$$s := (1 \quad -1 \quad 1 \quad \cdots \quad (-1)^{n-1})^T, \quad s_i = (-1)^{i-1}.$$

For any i, j , the integers $|i - j|$ and $i - j$ have the same parity, and $(i - j) - (i + j) = -2j$ is even. Consequently,

$$(-1)^{|i-j|} = (-1)^{i-j} = (-1)^{i+j} = s_i s_j.$$

Thus ss^T has the required off-diagonal entries and has diagonal entries $s_i^2 = 1$. It follows that

$$A = I_n + ss^T.$$

We now determine the eigenvalues of this rank-one perturbation. Let

$$W := \{x \in \mathbb{C}^n : s^T x = 0\}.$$

For every $x \in W$,

$$Ax = (I_n + ss^T)x = x + s(s^T x) = x,$$

so 1 is an eigenvalue on W . Since $s^T s = \sum_{i=1}^n s_i^2 = n$,

$$As = (I_n + ss^T)s = s + s(s^T s) = (n + 1)s,$$

so $n + 1$ is an eigenvalue with eigenvector s .

It remains to verify that these account for all n dimensions. Because $s^T s = n \neq 0$, every $x \in \mathbb{C}^n$ has the decomposition

$$x = \left(x - \frac{s^T x}{n} s \right) + \frac{s^T x}{n} s,$$

where the first summand lies in W . Also, $W \cap \text{span}\{s\} = \{0\}$: if $\alpha s \in W$, then $0 = s^T(\alpha s) = \alpha n$, hence $\alpha = 0$. Therefore

$$\mathbb{C}^n = W \oplus \text{span}\{s\}, \quad \dim W = n - 1.$$

Relative to a basis adapted to this direct sum, the matrix of A is $\text{diag}(1, \dots, 1, n + 1)$, with $n - 1$ copies of 1. Hence

$$\det A = 1^{n-1}(n + 1) = n + 1.$$